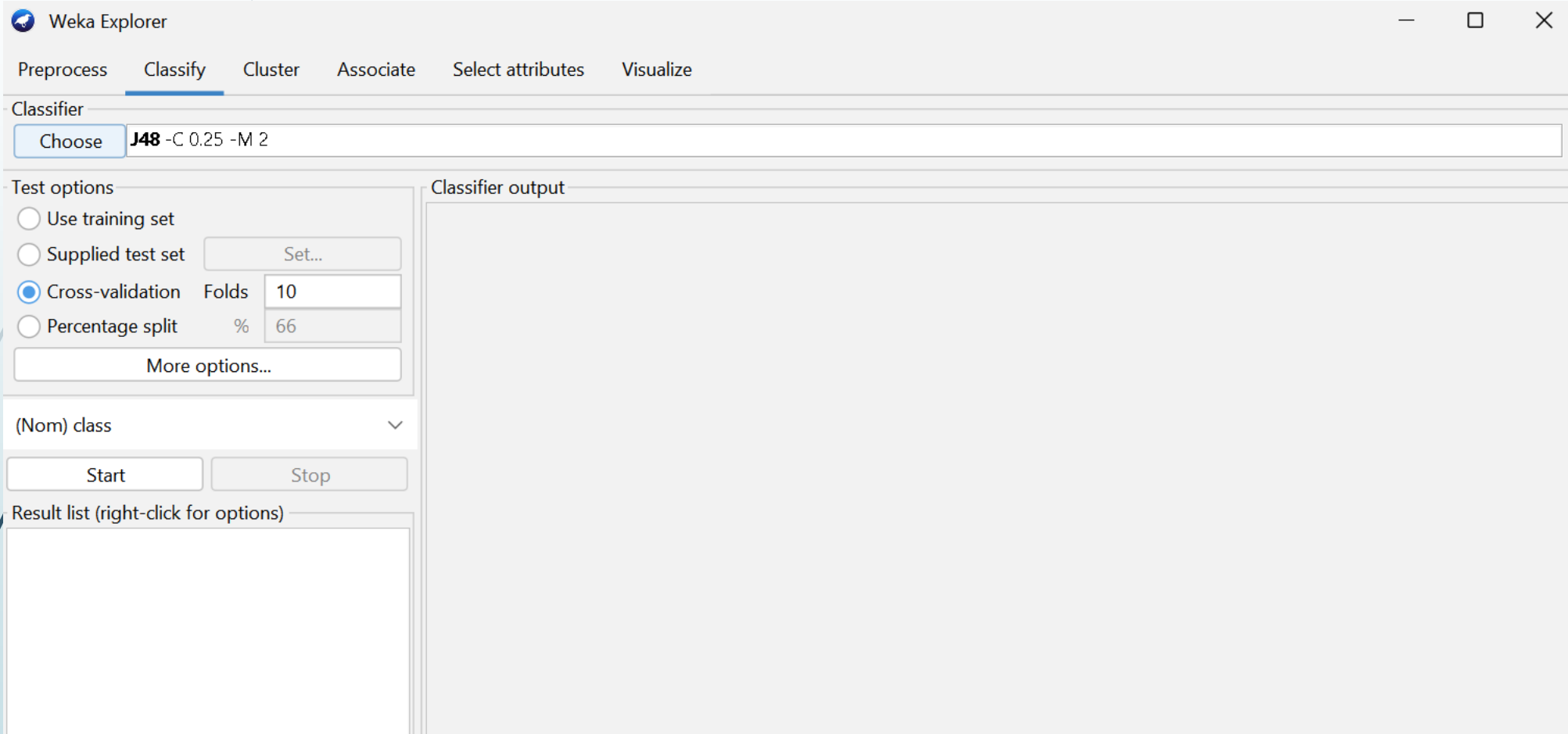


APLICACIÓN DE ALGORITMO J48 A IRIS EN WEKA

PROGRAMACIÓN AVANZADA

EMILIANO NARCISO PERALTA BARRIENTOS
DANIELA MAITE REYES ARENA



Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

J48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds

10

☐ Percentage split

%

66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

21:37:38 - trees.J48

Classifier output

=== Run information ===

Schema: weka.classifiers.trees.J48 -C 0.25 -M 2

Relation: iris

Instances: 150

Attributes: 5

sepalength

sepalwidth

petallength

petalwidth

class

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree

petalwidth <= 0.6: Iris-setosa (50.0)

petalwidth > 0.6

| petalwidth <= 1.7

| | petallength <= 4.9: Iris-versicolor (48.0/1.0)

| | petallength > 4.9

| | | petalwidth <= 1.5: Iris-virginica (3.0)

| | | petalwidth > 1.5: Iris-versicolor (3.0/1.0)

| petalwidth > 1.7: Iris-virginica (46.0/1.0)

Number of Leaves : 5

Size of the tree : 9

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

J48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds

10

☐ Percentage split

%

66

More options...

(Nom) class

▼

Start

Stop

Result list (right-click for options)

21:37:38 - trees.J48

Classifier output

Time taken to build model: 0.15 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	144	96	%
Incorrectly Classified Instances	6	4	%
Kappa statistic	0.94		
Mean absolute error	0.035		
Root mean squared error	0.1586		
Relative absolute error	7.8705 %		
Root relative squared error	33.6353 %		
Total Number of Instances	150		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.980	0.000	1.000	0.980	0.990	0.985	0.990	0.987	Iris-setosa
	0.940	0.030	0.940	0.940	0.940	0.910	0.952	0.880	Iris-versicolor
	0.960	0.030	0.941	0.960	0.950	0.925	0.961	0.905	Iris-virginica
Weighted Avg.	0.960	0.020	0.960	0.960	0.960	0.940	0.968	0.924	

=== Confusion Matrix ===

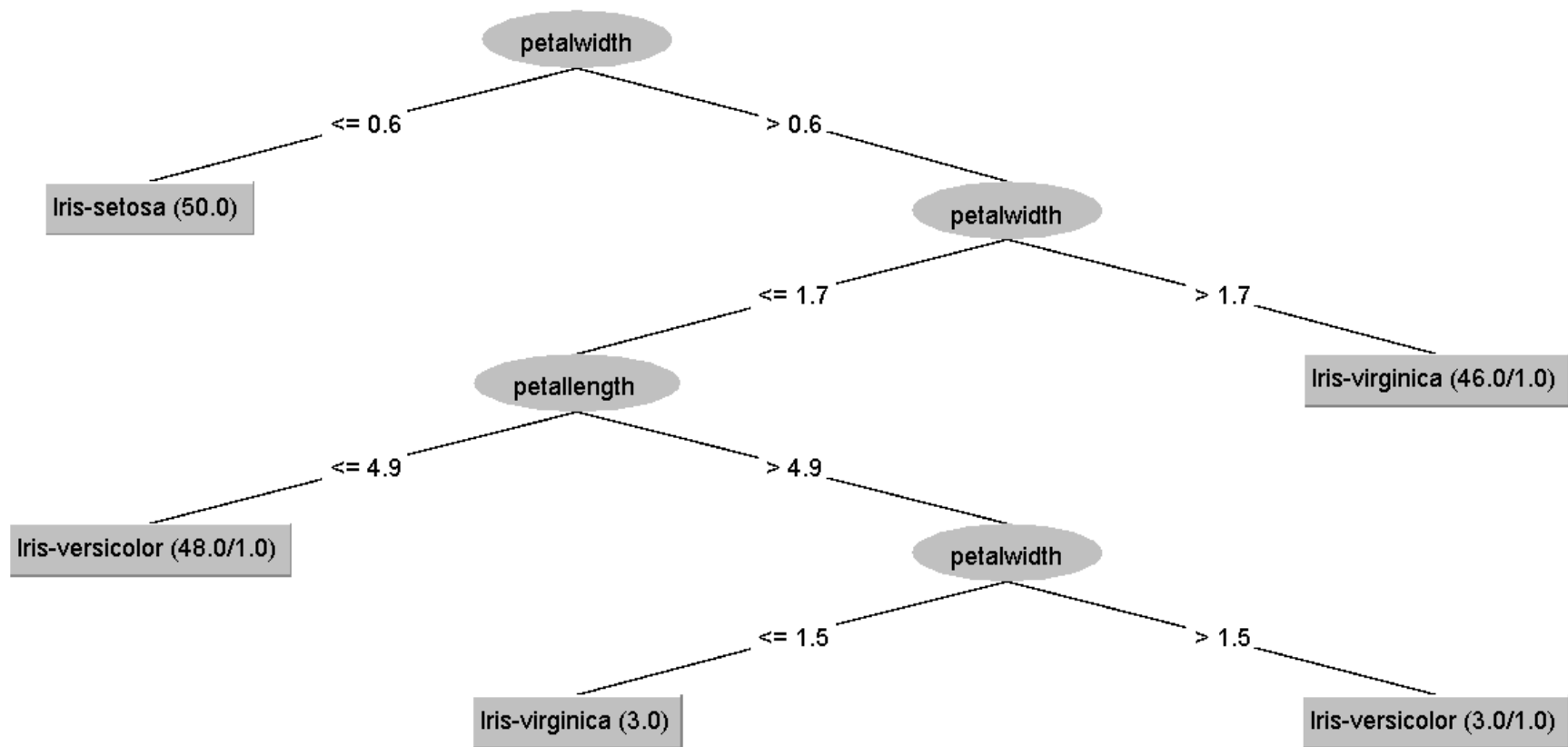
a b c

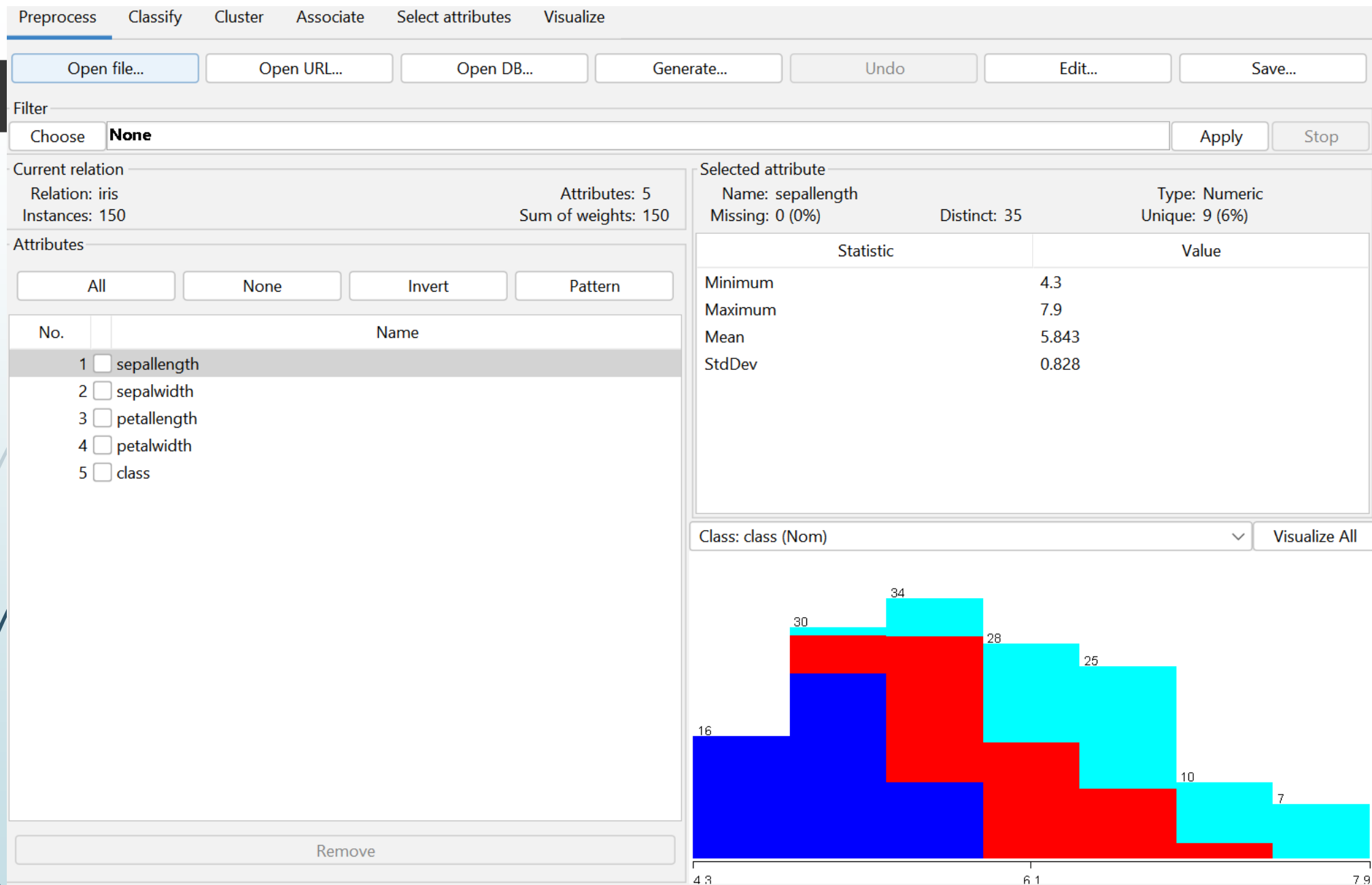
<-- classified as

49 1 0 | a = Iris-setosa

0 47 3 | b = Iris-versicolor

0 2 48 | c = Iris-virginica





Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseJ48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds10

☐ Percentage split

%66

More options...

(Nom) Class

StartStop

Result list (right-click for options)

21:37:38 - trees.J48

22:06:44 - trees.J48

Classifier output

=== Run information ===

Scheme:weka.classifiers.trees.J48 -C 0.25 -M 2

Relation:vote

Instances:435

Attributes:17

handicapped-infants

water-project-cost-sharing

adoption-of-the-budget-resolution

physician-fee-freeze

el-salvador-aid

religious-groups-in-schools

anti-satellite-test-ban

aid-to-nicaraguan-contras

mx-missile

immigration

synfuels-corporation-cutback

education-spending

superfund-right-to-sue

crime

duty-free-exports

export-administration-act-south-africa

Class

Test mode:10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree

physician-fee-freeze = n: democrat (253.41/3.75)

Classifier

Choose J48 -C 0.25 -M 2

Test options

- ☐ Use training set
- ☐ Supplied test set Set...
- ☒ Cross-validation Folds 10
- ☐ Percentage split % 66

More options...

(Nom) Class

Start

Stop

Result list (right-click for options)

21:37:38 - trees.J48

22:06:44 - trees.J48

Classifier output

```
-----
physician-fee-freeze = n: democrat (253.41/3.75)
physician-fee-freeze = y
|  synfuels-corporation-cutback = n: republican (145.71/4.0)
|  synfuels-corporation-cutback = y
|  |  mx-missile = n
|  |  |  adoption-of-the-budget-resolution = n: republican (22.61/3.32)
|  |  |  adoption-of-the-budget-resolution = y
|  |  |  |  anti-satellite-test-ban = n: democrat (5.04/0.02)
|  |  |  |  anti-satellite-test-ban = y: republican (2.21)
|  |  mx-missile = y: democrat (6.03/1.03)
```

Number of Leaves : 6

Size of the tree : 11

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	419	96.3218 %
Incorrectly Classified Instances	16	3.6782 %
Kappa statistic	0.9224	
Mean absolute error	0.0611	
Root mean squared error	0.1748	
Relative absolute error	12.887 %	
Root relative squared error	35.9085 %	
Total Number of Instances	435	

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

J48 -C 0.25 -M 2

Test options

Use training set

Supplied test set

Cross-validation

Folds

10

Percentage split

%

66

More options...

(Nom) Class

Start

Stop

Result list (right-click for options)

21:37:38 - trees.J48

22:06:44 - trees.J48

Classifier output

Size of the tree : 11

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances41996.3218 %

Incorrectly Classified Instances163.6782 %

Kappa statistic0.9224

Mean absolute error0.0611

Root mean squared error0.1748

Relative absolute error12.887 %

Root relative squared error35.9085 %

Total Number of Instances435

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.970	0.048	0.970	0.970	0.970	0.922	0.971	0.965	democr
	0.952	0.030	0.952	0.952	0.952	0.922	0.971	0.947	repub
Weighted Avg.	0.963	0.041	0.963	0.963	0.963	0.922	0.971	0.958	

=== Confusion Matrix ===

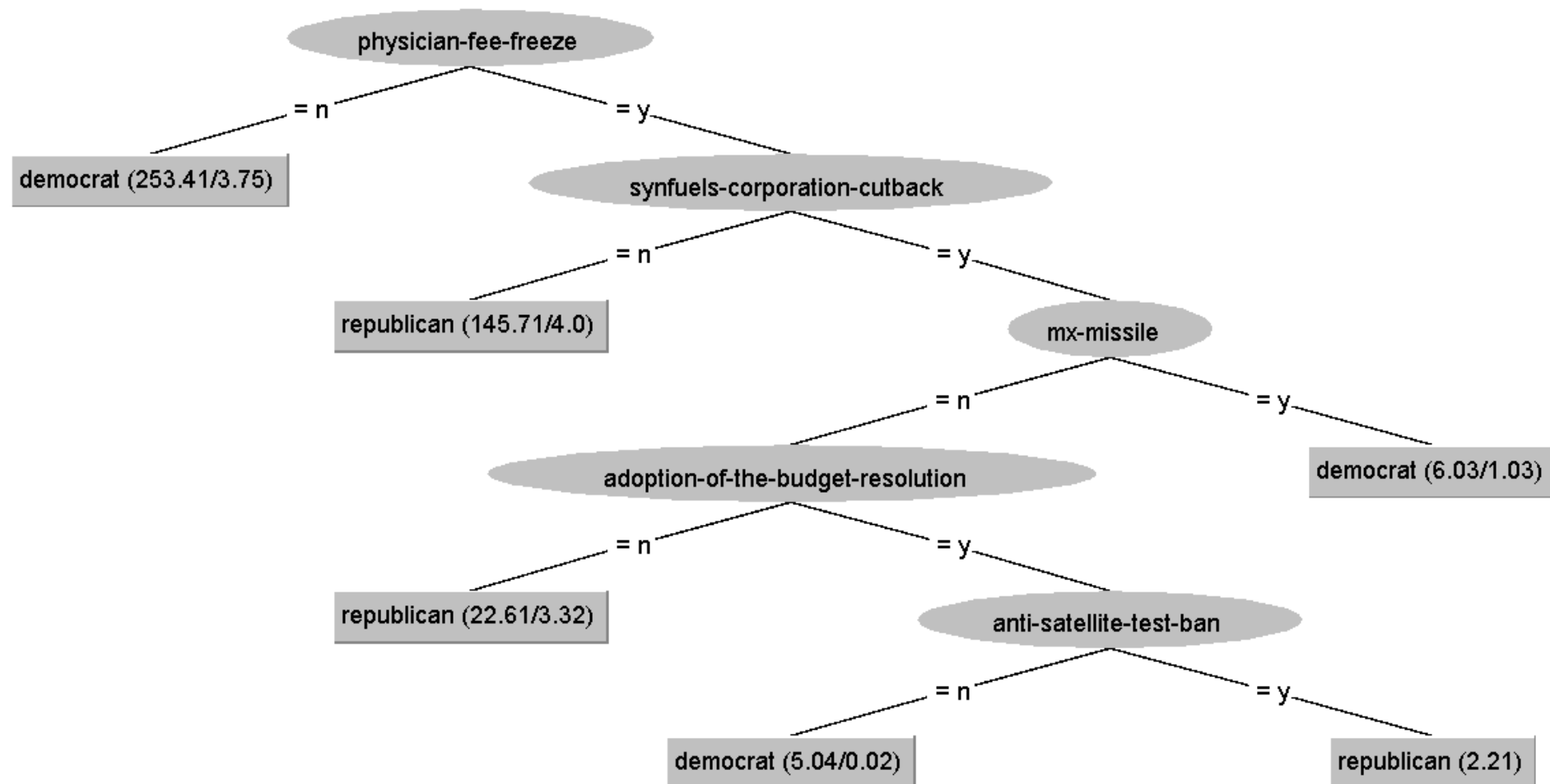
a b <-- classified as

259 8 | a = democrat

8 160 | b = republican



Tree View



Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

Filter

ChooseNone

ApplyStop

Current relation

Relation: vote

Instances: 435

Attributes: 17

Sum of weights: 435

Attributes

AllNoneInvertPattern

No.	Name
1	☒ handicapped-infants
2	☐ water-project-cost-sharing
3	☐ adoption-of-the-budget-resolution
4	☐ physician-fee-freeze
5	☐ el-salvador-aid
6	☐ religious-groups-in-schools
7	☐ anti-satellite-test-ban
8	☐ aid-to-nicaraguan-contras
9	☐ mx-missile
10	☐ immigration
11	☐ synfuels-corporation-cutback
12	☐ education-spending
13	☐ superfund-right-to-sue
14	☐ crime
15	☐ duty-free-exports
16	☐ export-administration-act-south-africa
17	☐ Class

Remove

Selected attribute

Name: handicapped-infants

Missing: 12 (3%)

Distinct: 2

Type: Nominal

Unique: 0 (0%)

No.	Label	Count	Weight
1	n	236	236
2	y	187	187

Class: Class (Nom)Visualize All

236

187



Mi experiencia con Weka fue muy práctica. Aprendí que su entorno visual facilita implementar árboles de decisión sin necesidad de programar, permitiéndome enfocarme en el análisis. Comprendí de forma clara cómo el modelo clasifica los datos mediante atributos clave y reglas lógicas, lo que hace mucho más sencilla la interpretación de resultados y la evaluación del algoritmo.